INCH-POUND

MIL-O-44109B
31 December 1990
SUPERSEDING
MIL-O-44109A
4 August 1987

MILITARY SPECIFICATION

OVERALLS, COMBAT VEHICLE CREWMEN'S

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers overalls made of fire-resistant material.

1.2 Classification. The overalls shall be of one type in the following sizes (see 6.2):

Schedule of sizes

Short	Regular	Long
X-Small	X-Small	X-Small
Small	Small	Small
Medium	Medium	Medium
Large	Large	Large
X-Large	X-Large	X-Large

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8415

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-50083	- Bag, Plastic, Folded Garment
A-A-50195	- Thread, Aramid
A-A-50199	- Thread, Polyester Core: Cotton- or Polyester-Covered
T-T-881	- Twine, Cotton, Seine
V-F-106	- Fastener, Slide, Interlocking
TT-E-529	- Enamel, Alkyd, Semigloss, Low Voc Content
DDD-L-20	- Label: For Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	- Boxes, Shipping, Fiberboard

MILITARY

MIL-E-20652/1	- Eyelets, Metallic, Rolled Flange Type; and Eyelet Washer
MIL-F-21840	- Fastener Tapes, Hook and Loop, Synthetic
MIL-L-35078	- Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage, General Specification For
MIL-C-43303	- Cord, Elastic, Cotton
MIL-T-43709	- Tape, Textile, Aramid
MIL-C-43842	- Cloth, Oxford, Aramid
MIL-B-81813	- Batting, Aramid or Novoloid Fiber, Quilted
MIL-T-83193	- Thread, Aramid, Spun Staple

STANDARDS

FEDERAL

FED-STD-595	- Colors Used in Government Procurements
FED-STD-751	- Stitches, Seams and Stitchings

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 633 - Electrodeposited Coatings of Zinc on Iron and Steel
D 3951 - Standard Practice for Commercial Packaging

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

THE COLOR ASSOCIATION OF THE UNITED STATES

Standard Color Card of America

Department of Defense Standard Shades for Sewing Threads

(Color cards may be available from the Color Association of the United States, Inc., 343 Lexington Avenue, New York, NY 10016-0927. If color cards are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3), in accordance with 4.3.

3.2 Guide sample. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Basic material. The basic material shall be 6.0 ounce aramid oxford cloth conforming to MIL-C-43842. The color shall be Olive Green 106.

3.3.2 Lining. The material for the lining shall be aramid or novoloid quilted batting conforming to MIL-B-81813. The face side of the lining material shall be Olive Green 106 and the stitching of dumbbell pattern shall have Olive Drab thread. The opposite side may be Olive Green 106, or Natural or Mixed (OG warp - natural filling or natural warp - OG filling) and the stitch of the dumbbell pattern shall have natural thread.

3.3.3 Cotton elastic webbing. The cotton elastic webbing for the elastic suspender straps shall be 1-15/32 inch wide conforming to Blackstone Webbing Company, Style No. 2662 or Style No. 2483 or equal (see 6.5). The color of the webbing shall match Olive Green 106.

3.3.4 Netting. The netting for the crotch inserts shall be aramid, tricot knit, 3.5 to 4.2 oz/sq yd, dyed OG-106 conforming to FAB Industries, Style No. SSM-605 or equal (see 6.5). The netting shall be fabricated from 200 denier, E. I. Dupont type 432 aramid yarn or equal (see 6.5) and shall have a minimum bursting strength of 140 pounds.

3.3.5 Fastener tapes. The hook and loop fasteners shall be Green 3421, 2 inches wide conforming to type I, class 2 of MIL-F-21840.

3.3.6 Cord. The cord beading used for welts of pocket openings and front openings shall conform to type I, size 18 of T-T-881.

3.3.7 Slide fasteners. The slide fasteners shall be brass with a black chemical finish and conform to size M of V-F-106 in the following types and styles:

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Type I, style 3 - Drop set, and leg side openings
Type I, style 7 - All pockets
Type I, style 15 - Front openings

3.3.7.1 Length of slide fasteners (excluding tape).

Front opening (1 each required)	Short - 22 inches Regular - 23 inches Long - 24 inches
Front pockets (2 required)	7-1/2 inches
Drop seat (2 each required)	Short - 16 inches Regular - 17 inches Long - 18 inches
Leg opening (2 required)	12 inches
Thigh pocket (2 required)	8-1/2 inches
Shin pocket (2 required)	8-1/2 inches

3.3.7.2 Slide fastener tapes. The material for the slide fastener tapes shall be aramid dyed to match the basic material. The tapes shall be of sufficient lengths to comply with respective slide fastener attachment operations. Color matching of the fastener tape under incandescent lamplight at 2300 ± 100 K shall be exempt.

3.3.7.2.1 Colorfastness. The finished tape shall show fastness to light and laundering equal to or better than the standard sample. When no standard sample is available, the tape shall show "fair" fastness to light after 6 standard fading hours, and "good" fastness to laundering.

3.3.7.3 Slide fastener thongs. The slide fastener thongs shall be 1/4 inch wide tape, Olive Green 106 conforming to MIL-T-43709. All thong tape ends shall be either heat cut or bartack sewn to prevent fraying.

3.3.8 Thread. The thread shall be aramid, Olive Green 483 matching color chip 34094 of FED-STD-595, sizes E and B conforming to type I of A-A-50195. Size E (90 tex) thread shall be used for the top thread of 401 stitching and size B (60 tex) for all other stitching. As an alternate, tex sizes 90 and 60 conforming to MIL-T-83193 may be used in lieu of the size E and B threads. The thread shall show colorfastness to laundering equal to the standard sample. When no standard sample is available, the thread shall show "good" fastness to laundering.

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3.3.8.1 Alternate cotton- and polyester-covered thread. Cotton- and polyester-covered core thread, conforming to ticket No. 50, 2 ply of A-A-50199 may be used as alternate thread for bartacking. The thread shall be dyed Olive Green 483 matching color chip 34094 of FED-STD-595, and shall show fastness to laundering equal to or better than the standard sample. When no standard sample is available, the thread shall show "good" fastness to laundering.

3.3.9 Belt slides. The suspender adjustment slides shall be 1-1/2 inches wide conforming to Waterbury Buckle Co., No. 95 slide, or North Judd No. 02300 or equal (see 6.5). The finish shall be zinc plate conforming to type IV, thickness class 13 of ASTM B 633 followed by baked enameling conforming to Class B of TT-E-529. The color shall be black.

3.3.10 Suspender loops. The size of the suspender loops shall be 1-1/2 inches. The loops shall conform to Scovill Mfg. Co. Howe or Peerless style or equal (see 6.5). The finish shall be zinc plate conforming to type IV, thickness class 13 of ASTM B 633 followed by baked enameling conforming to class B of TT-E-529. The color shall be black.

3.3.11 Button tack and rivet. The button tack shall conform to Scovill No. 74-3334-27-13-0161 or equal (see 6.5), and the rivet shall conform to Scovill No. 74-0533-14-01-1163 or equal (see 6.5). The rivet shall be copper plated. Both parts shall have a black chemical finish.

3.3.12 Eyelets. The eyelets for the leg cuff drawcord outlets shall conform to Part No. M206521/1 - BBE125 of MIL-E-20652/1, except that the size shall be 1/4 $\pm$ 1/32 inch inside diameter.

3.3.13 Leg cuff drawcord. The cord for leg cuff drawcords shall be a cotton elastic cord 3/16 of an inch in diameter, Olive Drab 7, conforming to class 2 of MIL-C-43303.

3.3.14 Labels. Each overall shall have an identification label, a size label, and an instruction label, conforming to type VI of DDD-L-20. The color of the label cloth shall be medium green (see 6.6). The fastness to laundering requirements of DDD-L-20 shall apply.

3.3.14.1 Identification label. The identification label shall conform to class 1.

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3.3.14.2 Size label. The size label shall conform to class 2 and shall contain the following information for each respective size. Descriptive information shall be printed in letters and numerals not less than 10-point and as follows:

X-Small - Short	X-Small - Regular	X-Small - Long
Height: Below 67 in. Stock No. NATO size: 6070/7484	Height: From 67 to 71 in. Stock No. NATO size: 7080/7484	Height: Over 71 in. Stock No. NATO size: 8090/7484
Small - Short	Small - Regular	Small - Long
Height: Below 67 in. Stock No. NATO size: 6070/8494	Height: From 67 to 71 in. Stock No. NATO size: 7080/8494	Height: Over 71 in. Stock No. NATO size: 8090/8494
Medium - Short	Medium - Regular	Medium - Long
Height: Below 67 in. Stock No. NATO size: 6070/9404	Height: From 67 to 71 in. Stock No. NATO size: 7080/9404	Height: Over 71 in. Stock No. NATO size: 8090/9404
Large - Short	Large - Regular	Large - Long
Height: Below 67 in. Stock No. NATO size: 6070/0414	Height: From 67 to 71 in. Stock No. NATO size: 7080/0414	Height: Over 71 in. Stock No. NATO size: 8090/0414
X-Large - Short	X-Large - Regular	X-Large - Long
Height: Below 67 in. Stock No. NATO size: 6070/1424	Height: From 67 to 71 in. Stock No. NATO size: 7080/1424	Height: Over 71 in. Stock No. NATO size: 8090/1424

3.3.14.3 Instruction label. The instruction label shall conform to class 3 and shall contain the following information:

INSTRUCTIONS

This overall is of low flammability. No other special flame resistant treatments are required.

1. Wear over coveralls for extra warmth.

2. Brush snow or frost from garments before entering heated shelters.

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3. For thorough cleaning, return to laundry unit for machine washing in accordance with established procedures. Overalls may be hand laundered; make sure that all soap is thoroughly rinsed out. The overall may be dry cleaned.

4. WARNING: Launder with slide fastener and hook and loop fasteners engaged. Do not press hook and loop fasteners. After 10 laundering cycles and thereafter, launder in accordance with Formula IV (no bleach) of FM 10-280, field laundry and Natick Formula X, fixed laundry for the retreatment of anti-static agent.

5. DO NOT STARCH. STARCH OR ANY OTHER TREATMENT THAT LEAVES A RESIDUE WILL DETERIORATE FLAME RESISTANCE.

DO NOT REMOVE THIS LABEL

3.3.14.4 Combination identification-instruction label. The identification label (class 1) and the instruction label (class 3) may be combined into one label. When combined, the item nomenclature shall not be repeated. The combination label shall conform to the requirements of the class 3 label.

3.4 Design. The overalls have a bib front with a center front slide fastener extending to the crotch, a drop seat, adjustable suspenders, and leg opening slide fasteners. The overalls have an integrated liner that extends above the waist level in the front and back. All pockets have slide fastener closures and are located at the left and right front hips, right and left upper thighs and right and left lower legs.

3.5 Patterns. Standard patterns which provide an allowance of 1/2 inch for all double-lapped and double-stitched seams, and 3/8 inch for all other seams, including liner joining seams, unless otherwise indicated, will be furnished by the Government to the contractor for cutting working patterns. The working patterns shall be duplicates of the Government patterns.

3.5.1 List of pattern parts. The pattern components shall be cut from the material specified and according to the pattern parts indicated.

List of pattern parts

Material	Pattern nomenclature	Cut parts
Basic material	Front	2
	Back	2
	Yoke, inner	1
	Yoke, outer	1

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List of pattern parts (cont'd)

Material	Pattern nomenclature	Cut parts
Basic material (cont'd)		
	Seat reinforcement	2
	Knee reinforcement	2
	Ankle reinforcement	2
	Zipper reinforcement	3
	Front pocket, inner	2
	Front pocket, outer	2
	Thigh pocket	2
	Shin pocket	2
	Side closure tab	4
Lining material	Back lining	2
	Front lining	2
Netting material	Crotch insert	4
Hook and loop tape	Hook tape	1

3.6 Construction.

3.6.1 Stitches, seams, and stitching. Stitch, seam, and stitching types specified in table I are those covered by FED-STD-751. When two or more seams or stitch types are given for the same part of an operation, any one of them may be used. When stitch type 401 is used, the looper thread shall be on the inside of the overall. Seam allowance shall be maintained with seams sewn so that no raw edges, run-offs, twists, pleats, puckers, or open seams will result. All seams shall start and finish evenly. Thread tension shall be maintained so that there is no tight or loose stitching. The gauge of stitching for double-lapped and double-stitched seams shall be 1/4 to 5/16 inch. The guides and knives on the overedge machines producing stitch type 501, 502, or 503 shall be set to trim only the ravelled edges of the fabric and in no case shall the amount trimmed be greater than 1/16 inch.

3.6.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in table I.

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3.6.3 Thread breaks and ends of seams. Ends of all seams and stitchings, when not caught in other seams and stitchings, shall be backtacked not less than 1/4 inch. Ends of a continuous line of stitching shall overlap. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than 1/2 inch. Skipped stitches or thread breaks on 401 stitch type may be repaired using 301 stitch type.

3.6.4 Bartacks. Bartacks shall measure 1/2 to 3/4 inch long, unless otherwise specified, and be free from thread breaks and loose stitching.

3.7 Manufacturing operations requirements. The overall shall be manufactured in accordance with operation requirements specified in table I. The contractor is not required to follow the exact sequence of operations listed. Any additional basting or holding stitching used to facilitate manufacture is permissible, provided the thread is removed or does not show on the finished overalls.

3.7.1 Figure. Figure 1 is furnished for information purposes only. To the extent of any inconsistencies between the document and figure 1, the document shall govern.

3.8 Use of automated apparel equipment. Automated apparel equipment may be used to perform any of the operations specified in table I, providing that the seam and stitch type are as specified and the finished component conforms to the required configuration. When a government furnished shaper pattern is forwarded, the component shall conform to that pattern.

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NO	TABLE I. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. Cut the overalls in strict accordance with the patterns furnished which show size, shape, directional lines (warp), placement of pockets and slide fastener openings or slash marks, reinforcement patches, cut out opening for leg opening, and indicators of seam allowances at edge of pattern. b. Cut all component parts of overall out of one piece of material for shell except the pockets and front zipper reinforcement piece, and underside of side closure tab, which may be cut from ends. Parts cut from ends shall approximate the shade of the overall. c. Cut the lining parts out of the same piece of material (care should be taken in cutting the batting material and handling the cut parts in that the ends of the quilting stitching having a tendency to run back). The face side of the lining parts (side with OD thread in dumbbell pattern) shall face the wearer (see 3.2.2). d. Cut the welt pieces for the slide fastener openings (except front and side closures and pockets) from the basic material on the bias, filling or warp direction 1-5/8 to 1-3/4 inches in width and shall extend the same length as the slide fastener chain (see 3.2.7), plus 1-1/2 to 2 inches. This will be the same length as the respective slide fastener tapes.						

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. (cont'd) e. The cord pieces for welt folded edges shall be cut to correspond to welt lengths cut in c. above. The cord pieces for the front opening shall be cut the length of the front top of bib to inseam minus 1/2 to 3/4 inch to allow for double-folded inseam. Cords for side openings shall be cut length of side zipper location notch from top of bib at side. f. Cut leg cuff drawcord sufficient length to allow for 3/4 inch beyond knot at both ends and so cord will not draw bottoms in when untied. Dip ends in wax to seal end. g. Cut suspender webbing, 24 ± 1/2 inches long for all groups and sizes. h. Cut the aramid netting for the crotch insert so as not to distort or twist. i. Cut the slide fastener pull thongs of sufficient lengths to finish 3 ± 1/4 inches when folded in half. Cut with hot knife. j. Cut lining binding 1-1/2 ± 1/4 inches wide and sufficient length to cover lining opening in operations. This may be precut and placed on rolls. Cut on bias of unquilted lining cloth.						

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NO	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. (cont'd) k. Cut two 1-1/4 inch square basic cloth for eyelet reinforcement patch in bias, filling, or warp direction. These may be cut from roll. l. Cut the hook and loop fastener tape pieces in the following widths and lengths: Use — Width (in.) — Length (in.) Side and center loop pieces on yoke — 2 — 10 Side closure loop — 2 — 4-7/8 Hook on side closure — 2 — 4						
2.	Lengths of slide fasteners (excluding tape and including top and bottom stops). Front opening (1) — Short - 22 inches; Regular - 23 inches; Long - 24 inches Front pocket (2) — 7-1/2 inches Drop seat (2) — Short - 16 inches; Regular - 17 inches; Long - 18 inches						

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NO	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
2.	Lengths of slide fasteners (excluding tape and including top and bottom stops). (cont'd) Leg opening (2) 12 inches Thigh pocket (2) 8-1/2 inches Shin pocket (2) 8-1/2 inches						
3.	Marking. a. Mark, ticket or bundle all component parts of the overalls to insure correct size, proper assembly, and uniform shade throughout the overalls. b. Any method of marking may be used except: 1. Metal fastening devices. 2. Sew-on tickets. 3. Adhesive type tickets which leave traces of adhesive on the material after removal of the tickets.						
4.	Replacement of defective components. During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in 4.4.2 shall be removed from production and replaced with nondefective and properly matched components.						

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TABLE I. (cont'd)
MANUFACTURING OPERATIONS REQUIREMENTS

NO		STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Attach thongs to slide fasteners (twelve each overall; see operation 1.i).						
	Thread thong through pull tab of slide fastener and with the ends even, single stitch or bartack the plies together through the center (1/16 inch off center tolerance) in the lengthwise direction. The top end of stitching shall finish 3/8 to 5/8 inch from the end of the metal pull tab. The bartack shall measure 1/4 to 3/8 inch in length.	301 or bartack	SSa-1	10-14 28 per bartack	B	B	
6.	Make side closure tabs.						
	a. Join plies of side closure tabs and stitch sides and points.	301 or 401	SSe-2(a)	10-14	B	B	
	b. Trim points, turn to finished position, stitch 1/8 to 3/16 inch from outer edge. The tab shall finish 2 ± 1/8 inches wide.	301	SSe-2(b)	10-14	B	B	
	c. Join hook portion of nylon fastener tape to tab so that one end is up to point where sides begin to taper. Stitch around edges of tape 1/16 to 1/8 inch from edge.	301	LSbj-1	10-14	B	B	
7.	Make yoke.						
	a. Position size label centered 1 to 1-1/4 inch from top raw edge of inner yoke and stitch on all four sides.	301	LSbj-1	10-14	B	B	
	b. Join inner and outer yoke pieces along four longest sides leaving the two shortest sides open.	301 or 401	SSe-2(a)	10-14	B	B	
	c. Trim top corners, turn open seam to inside, work out points and top stitch 1/4 to 5/16 inch from edge.	301	SSe-2(b)	10-14	B	B	

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NO	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Make yoke. (cont'd)						
	d. Join loop portions of nylon fastener tape to outer yoke (away from wearer) according to marks on pattern, stitching through all plies and with inner yoke smooth and free of distortions.	301	LSbj-1	10-14	B	B	
	e. Overedge stitch straight side edges with raw edge even.	502 or 503	SSa-1	6-8	E	E	
8.	Make suspender belts.						
	Thread elastic webbing through the belt slide and the suspender loop with right side of the elastic webbing up. The elastic webbing shall pass under the first bar of the slide, over the middle bar and under the final and third bar; through the appropriate opening of suspender loop and back around the middle bar of the slide with free end positioned between the two plies of the elastic webbing formed by the latter threading procedure. With a loop formed 1-3/4 to 2 inches beyond buckle, turn under raw edge 1/4 to 3/8 inch, and stitch through end of belt 1/8 inch from folded edge with a box X pattern 1/8 inch from edges and 5/8 to 3/4 inch wide.	301	EFb-2	6-8	B	B	
9.	Make welt pieces (see operation 1.d).						
	a. Fold welt pieces in half along the longest dimension; insert cord in fold, and single stitch 1/8 to 3/16 inch from folded edge.	301	SSbf-3(a)	10-14	B	B	

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NO	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Make welt pieces (see operation 1.d). (cont'd)						
	b. Overedge long raw edges of welt edges together.	502 or 503	SSa-1	6-8	E	E	
	NOTE: Welt pieces including cording may be cut in strips and placed on rolls; however, there shall be no piecing visible on the finished overalls.						
10.	Attach welt pieces to fastener tapes (see operation 2.).						
	Position welt pieces to the face side of respective fastener tapes so that cord edge is even with center of chain so finished cord edges will abut. Join the welts and slide fasteners together with a row of stitches 1/4 to 5/16 inch from back edges of each slide fastener tape.	301 or 401	SSa-1	10-14	B B	B E	
11.	Attach seat reinforcement pieces.						
	a. Position seat reinforcement pieces on right and left lower backs (as indicated on pattern), with straight raw edges even with raw edges of inseams and seat seams. Turn under straight edge of back 1/2 inch (at junction of seat and inseam) and stitch to back.	301	LSd-2	10-14	B	B	
	b. Fold under seam allowance of seat reinforcement and stitch with a double needle around outer curved edge to each lower back piece. The stitching may be continued along seat and inseams 1/16 and 1/8 inch from edges.	301	LSd-2	10-14	B	B	

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NO	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
12.	Join right and left back outer shells.						
	With right back (as worn) over left back, double lap and double stitch center back seam. Seat reinforcement patches should match at top and bottom ± 1/8 inch. Top back shall not be staggered more than 1/8 inch.	301 or 401	LSc-2	10-14	B E	B B	
13.	Make leg cuff - slide fastener openings.						
	a. Cut leg openings according to marks on the pattern forming tongue at each end. Cut width of tongue 3/4 inch and 1/2 inch long.						
	b. With right side of welt against right side of back leg, and back edge of welt even with raw edge of cut opening, stitch 3/8 inch from edge. Form rectangle at top end catching tongue. Top stops of slide fastener shall finish 3/8 to 1/2 inch below top opening. Bottom welt edge shall finish 3/8 to 1/2 inch above bottom raw edge of overall.	301 or 401 or 515, 516 or 517	LSq-2(a)	10-14 6-8	B B E	B E E	
	c. Bartack top end of each leg opening overedge stitching.	Bartack		28 per bartack	B	B	
	d. Overedge raw edges of welt, tape, and back leg opening together.	502 or 503	SSa-1	6-8	E	E	
	e. Turn welt and slide fastener to finished position, and with tapes flat and up against back side of opening, edge stitch 1/16 inch from folded edge of opening.	301 or 401	LSq-2(b)	10-14	B	B	

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
14.	Attach knee reinforcement pieces and make pleats.						
	a. Turn under top and bottom of knee reinforcement pieces 3/8 inch, and position to front legs according to marks on pattern. Wider part of knee piece is top. Raw edges of inseams of front and knee piece and outseams should match.						
	b. Stitch with double needle across top and bottom edges.	301	LSd-2	10-14	B	B	
	c. With raw edges even, match pleat notches front leg to knee piece. Fold pleats toward bottom of overalls.						
	d. Stitch 1/16 inch from all folded edges of pleats for a distance of 5/8 to 3/4 inch from raw edge and back tack.	301	OSe-1	10-14	B	B	
	e. Stitch a holding stitch along 1/8 to 1/4 inch from inseam and outseam raw edges extending 1/2 inch above and below knee piece.	301 or 401	SSa-1	10-14	B B	B B	
15.	Make front pocket openings, attach pockets.						
	a. Cut front pocket opening according to marks on pattern and cut a tongue at each end to measure 3/4 inch wide and 1/2 inch long.						
	b. With welt right sides against front and with back edge of fastener tape even with cut opening, stitch through all plies 3/8 inch from back edge. Top and bottom stops would be 3/8 inch to 1/2 inch	301 or 401	LSq-2(a)	10-14	B B	B B	

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
15.	Make front pocket openings, attach pockets. (cont'd)						
	from ends of opening. Form a rectangular finished opening catching tongues in stitching.						
	c. Overedge stitch raw edges of welt, tape, and front opening together along sides.	502 or 503	SSa-1	6-8	E	E	
	d. Turn welt and slide fastener through opening to finished position, and with tapes flat against back of opening, edge stitch 1/16 inch from folded edge overlapping beginning of stitching 1/2 inch.	301 or 401	LSq-2(b)	10-14	B B	B E	
	e. Position front inner pocket straight edge even with back of slide fastener tape and raw edge. Bottom of pocket slash shall be even with bottom welt opening. Face side of pocket should finish inside pocket.						
	f. Clip inner pocket at bottom of straight edge according to pattern (allowance is 1/4 inch for seam).						
	g. Stitch inner pocket, fastener tape, welt, and front cut opening edges together 1/8 to 1/4 inch from raw edge. Stitching shall not show on right side of garment. Option to stitch to bottom, turn at clip mark (1/4 inch to corner seam line) and continue to edge of pocket. If not continued to edge of pocket bartack at level of clipped seam line.	301 or 401	LSq-2(a)	10-14	B B	B E	

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NO	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
15.	Make front pocket openings, attach pockets. (cont'd)						
	h. Turn inner pocket into finished position, with bottom edge turned to inside against slide fastener tape 1/4 inch (if not caught in 15.g.), and with top raw edge of inner pocket turned under, edge stitch from top to bottom corner, turn and stitch to side of pocket. Edge stitch 1/16 inch from folded edge.	301	LSq-2(b)	10-14	B	B	
	i. Overedge side edge of outer front pocket.	502 or 503	SSa-1	6-8	E	E	
	j. Position outer pocket (ply facing wearer) to inner pocket, with curved edges even join with safety-stitch continuing to side edge of pocket overlapping overedge stitching. Bottom edge of welt and side of inner pocket should be caught in safety stitch.	515, 516 or 517	SSa-1	6-8	B E	E E	
	k. Turn pocket toward center front of overall with pocket and tapes flat, stitch from right side of garment across top and bottom width of welts, superimposing stitching over ends of welt opening edge stitching.	301	SSa-1	10-14	B	B	
	l. Bartack each end of pocket opening over edge stitching.	Bartack		28 per bartack	B	B	

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
16.	Make front opening.						
	Fold under front edge 3/4 inch, insert cord in fold to within 1/2 to 3/4 inch from bottom crotch seam, stitch along back edge of cord and continue stitching to bottom 1/8 inch from fold.	301	SSbf-3(a)	10-14	B	B	
17.	Make front and side zipper reinforcement and attach to zipper tapes.						
	a. Fold under sides of zipper reinforcement 1/4 inch and stitch 1/16 to 1/8 inch from fold.	301	OSf-1	10-14	B	B	
	b. Turn under top (wider) edge 1/4 inch. Position the reinforcement so that it overlaps bottom inside surface of the slide fastener tape, 1/4 ± 1/16 inch and is centered. Stitch 1/16 to 1/8 inch from fold.	301 or 401	LSb-1	10-14	B B	B E	
18.	Attach slide fastener to front and finish front seam.						
	Finished appearance. Finished corded edges shall abut at bottom opening without puckers or pulls.						
	a. Position slide fastener tapes under front welt so that center edge of chain is even with corded edge.						
	b. Stitch front to tape 1/4 inch back of cord stitching. Continue stitching through zipper reinforcement piece to bottom of fronts keeping reinforcement and bottom of tapes flat. Abut corded edges of fronts together and repeat operation for opposite side	301 or 401	SSa-1	10-14	B B	B E	

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NO	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
18.	Attach slide fastener to front and finish front seam. (cont'd)						
	assuring that corded edges abut at the crotch area, to form the front seam.						
	c. Reinforce front opening below slide fastener opening by superimposing a row of stitching over the stitchline back of corded edge (3/8 inch to 1/2 inch below bottom stop to bottom of fronts) backtacking. A continuous row of stitching is acceptable at bottom of slide fastener opening, continuing around bottom 1/8 to 1/4 inch from bottom crotch inseam ending at bottom of slide fastener opening, backtacking at both ends.	301	SSbf-3(a	10-14	B	B	
	d. Position bartack centered horizontally over abutted cord edges 3/8 to 1/2 inch below bottom stop of slide fastener catching top of stitching in operation 18.c.	Bartack		28 per bartack	B	B	
	e. Locate second bartack half way between bartack in operation 18.d. and bottom of fronts. Center bartack over corded edges, overlapping cord stitching.	Bartack		28 per bartack	B	B	
19.	Join front to back at inseam.						
	Position front over back, center back seat seam to front seam (± 1/8 inch) and match bottom leg (no ease). Raw edges of knee reinforcement and leg inseam should be even if holding stitch was not used.	301 or 401	LSc-2	10-14	B E	B B	

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
20.	Attach ankle reinforcement.						
	a. Fold under curved edges 3/8 inch, position ankle reinforcement pieces to inseam of leg matching top notch to center of inseam and bottom notch with bottom inseam and even with bottom legs.						
	b. Stitch with double needle (1/4 inch gauge) around curved edge.	301	LSd-2	10-14	B	B	
21.	Join front to back outseam below side opening.						
	a. Clip 1/2 inch to seamline at side slide fastener location on front and back.						
	b. Position front over back and stitch side seam assuring that pattern marks are aligned front to back.	301 or 401	LSc-2	10-14	B E	B B	
22.	Make side welts and attach slide fasteners.						
	a. Turn under 1/2 inch seam allowances from side slide fastener notch to top of side.						
	b. Insert a cord into the fold and stitch from the underside of its welt to top of side seam, with a row of stitching located 1/16 inch from the cord edge.	301	SSbf-3(a)	10-14	B	B	
	c. Position zipper reinforcement piece at side opening notch. Reinforce the side opening notch. Reinforce the side opening by superimposing two	301		10-14	B	B	

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NO.	TABLE I. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
22.	Make side welts and attach slide fasteners. (cont'd)						
	rows of stitching over the double lapped seam stitching for a distance of 1-1/4 inches. Position slide fastener face down on corded welt side of opening so that edge of chain is even with corded edge. Bottom stops should be even with side seam notch.						
	d. Stitch 1/4 inch from cord stitching from top to level of side seam on both sides and across bottom 3/4 inch catching zipper reinforcement piece. When finished and slide fastener is in closed position, corded edges should abut. Caution should be taken not to catch front pocket in stitching. Top side seams should not stagger more than 1/8 inch.	301 or 401	SSa-1	10-14	B B	B B	
	e. Bartack across top of side seam, catching abutted corded edges and zipper reinforcement piece. A second bartack shall be placed 1/2 to 5/8 inch below the first, crossing the side seam and catching the zipper reinforcement piece. Raw edges shall be caught and the slide fastener tapes and welt shall finish flat.	Bartack		28 per bartack	B	B	
23.	Make thigh and shin pocket openings and attach zippers and pockets.						
	a. Position wrong side of pocket to wrong side of overall leg, matching pocket cut location marks and stitch on either side of pocket cut location	301	OSa-1	10-14	B	B	

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TABLE I. (cont'd)
MANUFACTURING OPERATIONS REQUIREMENTS

NO.		STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
23.	Make thigh and shin pocket openings and attach zippers and pockets. (cont'd)						
	line. The row of stitches shall be located 1/8 ± 1/16 inch from the line. Stitching shall extend a minimum of 1/2 inch beyond each end.						
	b. Slash along pocket location line through leg and pocket ply and form tongues at ends 3/4 inch wide and 1/2 inch long.						
	c. Position back edge of slide fastener tape and welt to raw edge of pocket opening, and stitch 3/8 inch from back edge of fastener tape forming a rectangular opening 3/8 to 1/2 inch wide and catching tongues in stitching.	301	LSq-2(a)	10-14	B	B	
	d. Turn overalls into finished position and with tapes, welts and seam allowances flat, edgestitch 1/16 to 1/8 inch from folded edge. Overlap edgestitching beginning and ending across one end of opening.	301	LSq-2(b)	10-14	B	B	
	e. Bartack end of each pocket opening over edge stitching.	Bartack		28 per bartack	B	B	
24.	Make thigh and shin pocket pleats and close pockets.						
	a. Fold bottom pleats of pockets so centerfold is between pocket sides and with sides and tops even, stitch sides and top.	515, 516 or 517	SSa-1	6-8	B E	E E	
	or						

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					NEEDLE	BOBBIN/ LOOPER	COVER
24.	Make thigh and shin pocket pleats and close pockets. (cont'd)						
	b. Overedge stich raw edges of thigh and shin pockets.	502 or 503	SSa-1	6-8	E	E	
	c. Fold bottom pleats of pockets so centerfold is between pocket sides and with sides and tops even, stitch sides and top.	301 or 401	SSa-1	10-14	B E	B B	
25.	Make bottom leg opening in lining.						
	Stitch binding tape to side leg opening with a row of stitching located 1/16 to 1/8 inch from the back edge of the folded binding tape. The binding tape shall finish 1/2 ± 1/16 inches wide.	301	BSc-1	10-14	B	B	
26.	Attach label to front lining.						
	Position identification label 5-1/2 to 6 inches from top raw edge of left front (as worn) and 3/4 to 1 inch from front opening edge. Label should not interfere with curved side seam and stitch on all four sides. Stitching shall not be through printing. Position the instruction label immediately below the identification label. If the combination identification-instruction label is used, position it as stated above.	301	LSbj-1	10-14	B	B	
27.	Bind front pocket openings.						
	a. Slash along pocket cut out location according to pattern marks.						

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TABLE I. (cont'd)
MANUFACTURING OPERATIONS REQUIREMENTS

NO.		STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
27.	Bind front pocket openings. (cont'd)						
	b. Bind the raw edges in accordance with the procedure described in operation 25.						
	c. Trim ends of binding even with side of lining fronts. This operation could be omitted if side lining openings are trimmed with knife when overedged.						
28.	Join inseams of lining.						
	a. Fold pleats according to notches on pattern so that fold facing wearer is toward bottom of overall.						
	b. Option may be selected to stitch 3/8 to 1/2 inch from inseam across pleat to hold pleats prior to inseam.	301 or 401	SSa-1	10-14	B	B	
	c. Join front and back lining legs at inseam and stitch catching folded pleats.	515, 516 or 517	LSq-2(a)	6-8	B E	E E	
	d. Turn seam allowances against front leg and stitch 3/16 to 1/4 inch from first row.	301 or 401	LSq-2(b)	10-14	B B	B E	
29.	Join outseams of lining.						
	Join according to operations 28, a through d, up to the drop seat opening.						
30.	Attach crotch insert.						
	a. With double ply of inserts together, edges matching, align curved edge to curved edges of front/back lining crotch opening. Long point of crotch insert matching back.						

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					NEEDLE	BOBBIN/ LOOPER	COVER
30.	Attach crotch insert. (cont'd)						
	b. Stitch insert to crotch.	515, 516, or 517	LSq-2(a)	6-8	B E	E E	
	c. Turn seam allowance toward crotch inserts and stitch 3/16 to 1/4 inch from previous row.	301 or 401	LSq-2(b)	10-14	B B	B E	
31.	Close front pocket opening bottoms and bind upper side lining. With pocket ends together, bind drop seat opening in accordance with procedure described by operation 27.						
32.	Join front, back, and seat seam.						
	a. With the face sides together, join right and left pieces together at the back and crotch. Crotch inserts should match ± 1/8 inch.	515, 516, or 517	LSq-2(a)	6-8	B E	E E	
	b. Turn seam allowance to one side and stitch 3/16 to 1/4 inch from folded edge.	301 or 401	LSq-2(b)	10-14	B B	B E	
33.	Bind front opening. Bind front opening according to operation 27.						
34.	Attach lining to shell top bib.						
	a. Position top of lining and shell with right side to right side, with top and curved side edges evenly aligned. Join the two pieces at the top on curved edges.	515, 516, or 517	LSq-2(a)	6-8	B E	E E	

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					NEEDLE	BOBBIN/ LOOPER	COVER
34.	Attach lining to shell top bib. (cont'd)						
	b. Trim top corners, turn lining to inside working out edges and points; and top stitch through fronts 3/8 inch from folded edge. Top welt cord ends and side opening welt cord ends should be pulled between seam and caught in stitching.	301 or 401	LSq-2(b)	10-14	B B	B B	
	c. Position hook fastener tape to back top bib according to marks on pattern. Stitch around all edges, overlapping on side of fastener tape.	301	LSbj-1	10-14	B	B	
35.	Attach lining and shell at drop seat slide fasteners.						
	a. Position binding edges to back of slide fastener tape 3/8 inch from the chain and so that the bottom of binding opening is 3/8 inch to 1/2 inch from bottom fastener stop and 3/8 inch wide across the bottom.						
	b. Stitch around the opening through all plies with a row of stitching located 5/16 to 1/4 inch from outer row of stitching that joins the slide fastener to the outer shell. Stitching at top edge should end at row of stitching from operation 34.b and be back-tacked for reinforcement.	301		10-14	B	B	
36.	Attach lining front closures to shell.						
	Stitch the two plies together in accordance with procedure described by the previous operation.						

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					NEEDLE	BOBBIN/ LOOPER	COVER
37	Attach lining at leg bottoms and hem. a. Pull lining legs into their finished position with the bottom of the lining even with the hem notch, i.e., 1-3/8 inches from bottom of overalls. b. Position edge of binding 3/8 inch back of chain and stitch through all plies from bottom raw edge of lining to 5 inches above and backtack. The row of stitching shall be located 1/4 to 5/16 inch from the edge stitching on the outside of the leg opening.	301		10-14	B	B	
	c. Attach eyelets in accordance with pattern marks. Insert a reinforcement piece cut 1-1/4 inches square from the basic cloth at the under side prior to installing the eyelet. The eyelet shall not overlay outseam on back leg and bound edge of lining on front leg. d. Thread drawcord through eyelet with ends toward lining side of overall and sufficient length so leg bottoms will remain ungathered. Knot ends of drawcords so there is 3/4 inch extension of drawcord beyond knots. e. With drawcord in place, fold up hem of shell over raw edge of lining along hemline, turn under top edge 3/8 inch, pull welt corded bottom to inside forming angle, and stitch 1/16 to 1/8 inch from turned edge backtacking along angle and continuing around bottom. Care should be taken not to catch drawcord. The hem	301	SSe-1	10-14	B	B	

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					NEEDLE	BOBBIN/ LOOPER	COVER
38.	Attach loop fastener tape to front.						
	a. Position loop fastener tape according to marks on pattern.						
	b. Stitch through all plies around all four edges overlapping ends of stitching.	301	LSbj-1	10-14	B	B	
39.	Attach side closure tab.						
	a. Position raw edge of tab on upper back (as indicated on pattern), and with point facing center back, stitch the width of the tab 1/8 to 3/16 inch from raw edge of tab through all plies.	301	LSg-2(a)	10-14	B	B	
	b. Turn tab to finished position and stitch the width of the tab with a rectangular box stitch 3/16 to 1/4 inch wide. The row of stitching near the turned edge of the tab shall finish 1/16 inch from the folded edge.	301	LSg-2(b)	10-14	B	B	
40.	Attach yoke.						
	a. Position straight sides of yoke 1/4 to 3/8 inch below curved armhole of bib top, with straight edge in line with back edge of binding, and with outer yoke against lining.						
	b. Stitch yoke to overalls through all layers 1/8 to 1/4 inch from back edge of yoke. The stitching shall start at the bib top, continue the width of the yoke, and be backtacked at each end. Turn yoke towards bib front and raise stitch the width of the yoke 1/8 to 1/4 inch from turned edge, backtacking ends of stitching.	301	LSq-2	10-14	B	B	

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					NEEDLE	BOBBIN/ LOOPER	COVER
41.	Attach suspender belts, button tacks, and rivets to yoke. a. Turn under belt end 1/4 to 3/8 inch with wrong side of belt against outer yoke position in accordance with marks on pattern. Stitch the belt to the yoke with a 1 inch by 1-1/8 ± 1/8 inches rectangular box with an X pattern. The ends of the stitching should be overlapped 1/2 inch and the stitching should not be more than 1/8 inch from the edges of the webbing. The side edges of the belt should be parallel to and not more than 1/8 inch from the side of the bib. b. Attach tack to lining side of bib and button to shell side of bib top (according to pattern location marks).						
42.	Clean threads. Trim all thread ends and remove loose threads.						
43.	Prepare for packing. a. Remove all spots, stains, and shade or size markings. b. Close all slide fasteners and fastener tapes.						

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3.9 Finished requirements. The overalls shall conform to the finished measurements shown in table II.

TABLE II. Finished measurements (inches)

	X-Small	Small	Medium	Large	X-Large	Tolerance
1/2 waist 1/	19-3/4	21-3/4	23-3/4	25-3/4	27-3/4	± 3/4
Inseams 2/						
Short	28-1/2	28-1/2	28-1/2	28-1/2	28-1/2	± 1/2
Regular	30-1/2	30-1/2	30-1/2	30-1/2	30-1/2	± 1/2
Long	32-1/2	32-1/2	32-1/2	32-1/2	32-1/2	± 1/2
Outseams 3/						
Short	43-3/4	43-3/4	43-3/4	43-3/4	43-3/4	± 3/4
Regular	46-3/4	46-3/4	46-3/4	46-3/4	46-3/4	± 3/4
Long	49-3/4	49-3/4	49-3/4	49-3/4	49-3/4	± 3/4
1/2 Knee 4/ (all lengths)	11-1/2	12	12-1/2	13	13-1/2	± 1/2
1/2 Bottom 5/ (all lengths)	11	11	11	11	11	± 1/2
Suspender strap 6/	18	18	18	18	18	± 1/2

1/ Half waist measurement taken from folded edge to folded edge through the center of loop fastener tapes for side tabs and with drop seat and front zippers closed.

2/ Inseam is measured from the center of crotch seam to the bottom edge of leg along the seam.

3/ Outseam is measured from the top edge of the seam to the bottom of the overalls with the drop seat zipper closed.

4/ Half knee measurement shall be taken at the center of the knee patch from folded edge to folded edge.

5/ Half bottom measurement shall be taken at bottom edge of overall from folded edge to folded edge.

6/ The finished length of the suspender strap shall be measured from the base of the strap where it is joined to the back yoke to the front folded edge of the suspender strap in the suspender loop. The slide shall be adjusted to the maximum length and the suspender elastic measured in a relaxed state.

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3.10 Workmanship. The overalls shall conform to the quality of product established by this specification and occurrence of defects shall not exceed the applicable acceptance quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 End item visual examination. The end items shall be examined for the defects listed in table III. The lot size shall be expressed in units of overalls. The sample unit shall be one overall. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects, 15.0 for major and minor A combined defects and 40 for total (major, minor A and minor B combined) defects.

TABLE III. End item visual defects

Examine	Defect	Classification Major	Minor A	Minor B
Material defects and damages	Hole, cut, tear, or needle chew on outside:			
	- up to 1/8 inch in diameter or length			301
	- more than 1/8 up to 1/4 inch inclusive		201	
	- more than 1/4 inch	101		
	Hole, cut, tear or needle chew on inside:			
	- up to 1/4 inch in diameter or length			302
	- more than 1/4 up to 1/2 inch inclusive		202	
	- more than 1/2 inch	102		
	Any mend, darn, or patch on outside	103		
	Mend, darn or patch on inside (except on seam allowance):			
	- 1/4 up to 1/2 inch in diameter or length			303
	- more than 1/2 inch		203	
	Friction burn:			
	- when fibers are broken use "hole, cut, tear, or needle chew" classification			
	Drill hole exposed on outside			304
	Weakening defects:			
	Smash	104		
	Multiple float or loose slub:			
	- on outside		204	
	- on inside			305

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Material defects and damages (cont'd)	Any slub greater than an F on the Sears Fabric Defect Scale (see 6.7)	205		
	Shade bar:			
	- on outside	206		
	- on inside			306
Shaded parts	Any outside part shaded (except those cut from ends)	207		
Cleanness	Spot or stain on outside 1/4 inch or more (largest dimension in any direction)			307
	Five or more thread ends not trimmed to 1/2 inch or less on outside or nine or more thread ends not trimmed to 1/2 inch or less on the inside			308
	Two or more shade or size tickets not removed			309
	More than 10 loose threads of 1 inch or more throughout (inside and outside)			310
Component and assembly	Any component part or required operation omitted (unless otherwise classified herein)	208		
	Any component not as specified	209		
	Any component part twisted, distorted, pleated, misshaped, tight or full (unless otherwise classified herein)			311
	Any operation not as specified (unless otherwise classified herein)			312
Cutting	Any component part not cut in conformance with directional lines indicated on patterns or not in accordance with document requirements			313
Seams and stitching	Accuracy of seaming:			
	- Seam irregular, twisted, or pleated or wavy			314
	- Seam puckered (score only when on major portion of seam)			315

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)	Accuracy of seaming: (cont'd):			
	- Any part of overall caught in an unrelated operation or stitching		210	
	- End of stitching, when not caught in other seams or stitching, backtacked less than 1/4 inch			316
	- Thread break (all stitch types) stitched less than 1/2 inch beyond each end of break			317
	- Ends of a continuous line of stitching overlapped less than 1/2 inch (except label stitching)			318
	- Ends of stitching on label overlapped with less than three stitches			319
	Gage of stitching and seam allowance:			
	- Irregular, i.e., unevenly gaged, or corresponding stitchings not uniformly gaged (to be scored only when condition exists on more than 1/2 the length of the seam)			320
	- Width not as specified or not within range of gage specified or varies more than 1/16 inch when no range is specified			321
	- Edge of raised stitching sewn too close to edge resulting in damage to cloth		211	
	- Seam allowance not as specified or varies more than 1/8 inch (to be scored only when condition exists on more than 1/2 the length of the seam)		212	
	Open seam (all seams):			
	- Up to 1/2 inch inclusive			322
	- More than 1/2 up to 3/4 inch inclusive		213	
	- More than 3/4 inch	105		

NOTE: One or more broken stitches or two or more continuous skipped or runoff stitches on joining seam constitute an open seam. On double-stitched seams, a seam is considered open when one or both sides of seam is open.

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)	Raw edges on outside:			
	- Up to 1/2 inch inclusive			323
	- More than 1/2 up to 3/4 inch inclusive		214	
	- More than 3/4 inch	106		
	Raw edges on inside (except overedge edge) more than 1 inch			324
	Raw edges on overedged edge:			
	- 1 inch to 2 inches inclusive			325
	- More than 2 up to 3 inches inclusive		215	
	- More than 3 inches	107		
	Raw edges exposed under side closure tab:			
	- 1/4 to 1 inch inclusive			326
	- More than 1 inch		216	
	Raw edge on inside or outside of overalls caused by excessive seam allowance protruding from a double-lapped, double-stitched seam			327
	Run-offs:			
	Edge or raised stitching (when not resulting in an open seam):			
	- 1/4 to 1/2 inch inclusive			328
	- More than 1/2 inch		217	
	Seam and stitch type:			
	- Not specified seam or stitch type		218	
	- Looper thread on outside (when 401 stitch type is used)	108		
	- Any line of stitching omitted	109		
	Stitch tension:			
	- Loose tension, resulting in a loose seam:			
	- Up to 1/2 inch inclusive			329
	- More than 1/2 up to 3/4 inch inclusive		219	
	- More than 3/4 inch	110		
	- Loose tension on raised or edge stitching resulting in exposed loose thread			330

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)	Stitch tension: (cont'd)			
	- Tight tension (stitches break when normal strain is applied in the direction of the seam or stitching)	111		
	Stitches skipped or broken:			
	- 1/4 to 1/2 inch inclusive			331
	- More than 1/2 inch		220	
	Stitches per inch (to be scored only when condition exists on major portion of seam):			
	- All seams, except stitching of labels and overedge stitching:			
	- Less than the minimum		221	
	- More than the maximum			332
	- Stitching of label and overedge stitching, two or more stitches less than specified			333
Bartack or tacks	Missing, insecure, misplaced, not specified size or type, or not serving intended purpose		222	
Front, side or leg and pocket openings	Top stops more than 1/4 inch from top of overall or side and front opening		223	
	Corded edges not abutted when fastener is closed. NOTE: Corded edges overlapping or lapping less than 1/16 inch shall not be scored as a defect.		224	
	Ends of tapes and welts not caught in the line of stitching at the tongue notched ends of openings		225	
	Cord omitted from slide fastener coverings	112		
	Binding or lining less than 1/8 inch or more than 1/2 inch away from slide fastener chain, except where otherwise specified		226	
	Piecing of welt pieces visible at one or more slide fastener openings			334

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Front, side or leg and pocket openings (cont'd)	Cording at folded edges not extending the full length of slide fastener opening		227	
Inseams and outseams	Fronts not lapping back			335
	Outside fold of knee pleats not toward bottom			336
Leg Openings	Elastic cord caught in the hemming operation		228	
	Width of hem not as specified or irregular by more than 1/4 inch		229	
	Ends of elastic cord unknotted			337
Labels	Missing, incorrect or illegible:			
	- Size	113		
	- Identification, instruction or combination where applicable		230	
	Not stitched on all four sides		231	
	Stitching through the printing		232	
	Size label off center by more than 1/2 inch or any label not positioned as specified		233	
Seat reinforcement patch	Patches out of alignment at top by more than 1/8 inch		234	
Slide fastener	One or more defective, affecting function	114		
	Not type, size, or style specified		235	
Slide fastener thongs	One or more omitted	115		
	Less than 2-3/4 inches long or more than 3-1/4 inches long		236	
	Line of stitching or bartack joining ends, off center by more than 1/6 inch, or the top end of the stitching finishing less than 3/8 inch or more than 5/8 inch from end of metal pull tab			338

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TABLE III. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Slide closure tab	Side closure tab finishes more than 2-1/8 inches wide or less than 1-7/8 inches wide			339
	Hook extending more than 1/16 inch beyond edges of tab			340
Hook and loop fastener tape	Hook or loop flattened or missing from any tape area less than 25 percent		237	
	Hook or loop flattened or missing from any tape area more than 25 percent	116		
	Fastener tape not specified size		238	
	Fastener tape missing or misplaced not serving intended purpose	117		
Drop seat	Hook portion positioned on outside of yoke and the loop portion positioned on inside top edge of back bib		239	

4.4.3 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified in table II. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of overalls. The sample unit shall be one overall. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

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Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1. Level A preservation. Each pair of overalls, with suspenders clipped and legs fully extended and with slide fasteners closed, shall be laid flat, front down. Fold the leg bottoms up approximately 14 inches. Next bring the leg portion up even with the top of back bib and fold the clippped suspender straps across the back. Each completely folded overall shall measure approximately 24 inches by 15 inches. Each folded overall shall be individually unit packed in a close-fitting bag conforming to A-A-50083.

5.1.2. Commercial preservation. The overalls shall be preserved in accordance with ASTM D 3951.

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5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1. Level A packing. Six overalls of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each fiberboard container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed flat, six in depth within a shipping container. Inside dimensions of each shipping container shall be approximate 24 inches in length, 15 inches in width, and 12 inches in depth. Approximate dimensions are furnished as a guide only. Each fiberboard container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Six overalls of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each fiberboard container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A unit packs shall be packed flat, six in depth within a shipping container. Inside dimensions of each shipping container shall approximate 24 inches in length, 15 inches in width, and 12 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.2.3. Commercial packing. Overalls, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), overalls, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with the bonding means C and D or film bonding F or G. Pallet pattern shall be number 3 in accordance with the appendix of MIL-STD-147.

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5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951 as applicable.

5.4.1. Unit packs. Bagged unit packs shall have the required identification information legibly printed or stamped in black directly on the bag across the center face or on a white label inserted within the bag so as to permit ready identification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The overalls are intended for wear by combat vehicle crewmen of the Department of Defense. The overalls are intended to be worn, over the combat vehicles crewmen's coveralls to provide additional cold weather, environmental, and flame protection.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

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6.5 Equal item. Prior to the use of an "or equal" item, the contractor shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval by the responsible military agency.

6.6 Color of label. The following cable numbers of the standard Color Card of America are furnished for information and guidance as to the intensity of the shade of "Medium Green" desired for the labels specified:

Cable No. 70034
Cable No. 70130
Cable No. 70131

6.7 Fabric defect scales. Sears Fabric Defect Replica Kits are available from:

Sears Roebuck and Company
Department 817, (ATTN BSC 23-29)
Sears Tower
Chicago, IL 60684

6.8 Subject term (key word) listing.

Cold Weather Protection
Flame Protection

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:

Army - GL
Navy - NU
Air Force - 99

Review activities:

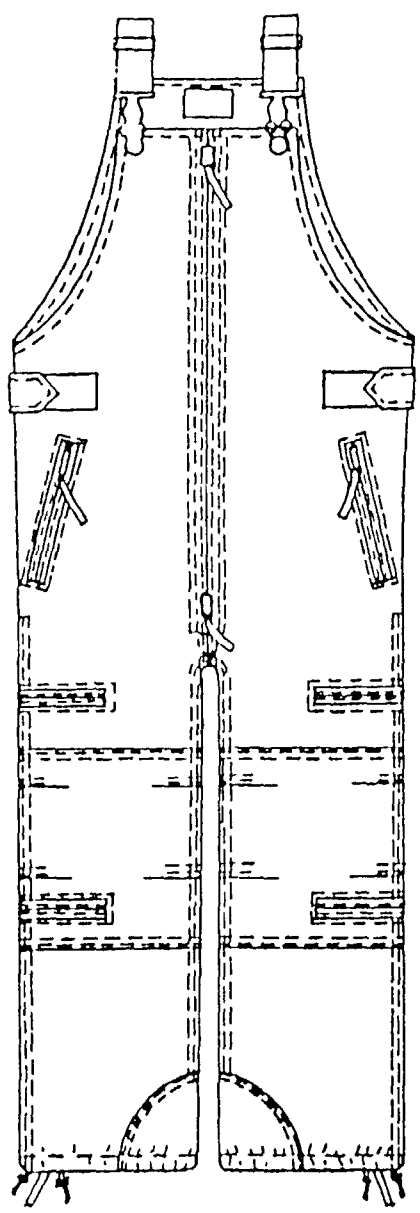
Army - MD
Navy - MC
Air Force - 82
DLA - CT

Preparing activity:

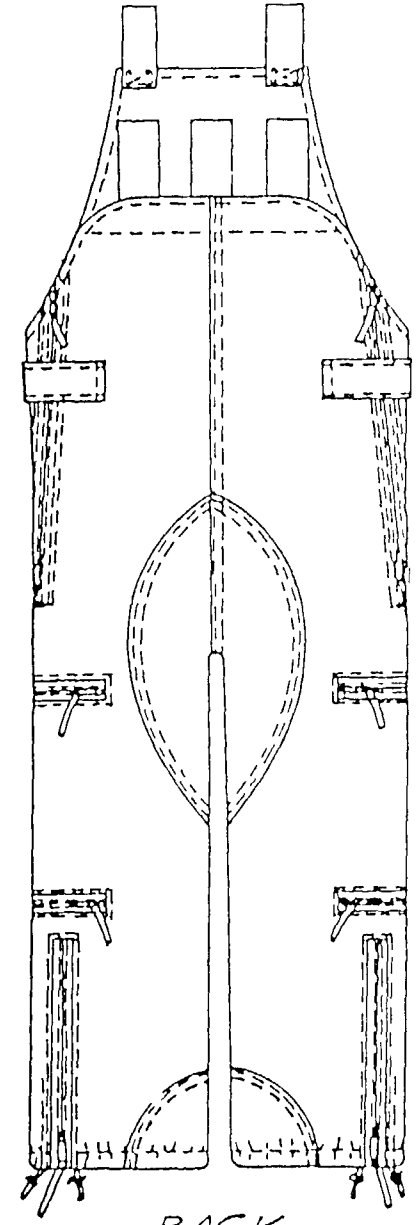
Army - GL

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FRONT



BACK

FIG. 1 OVERALLS, COMBAT VEHICLE CREWMEN'S

2-1-1546

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

NOTE. This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2 DOCUMENT DATE (YYMMDD)

3 DOCUMENT TITLE
OVERALLS, COMBAT VEHICLE CREWMEN'S

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
U S. Army Natick RD&E Center	508-651-4532	256-4532
c ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN STRNC-IRT Natick, MA 01760-5019	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT. Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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